

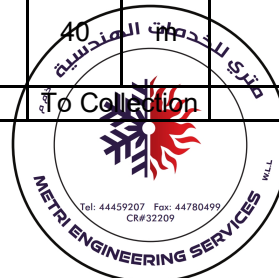
مركز للخدمات الهندسية

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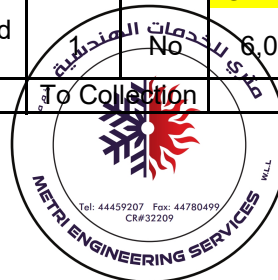
METRI ENGINEERING SERVICES

Tel: 44459207 Fax: 44780499
CR#32209

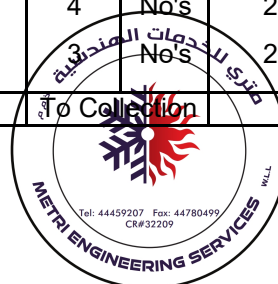
Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
A	<u>DX HI WALL UNITS</u> <u>Supply Installation, testing & Commissioning of Split DX Air conditioning Unit with Hi-wall mounted decorative type indoor unit and air cooled remote out door unit, R407C refrigerant, insulated refrigerant pipe works, condensate drain pipe and electrical connections, remote control etc. as per drawing, equipment schedule and technical specification of following sizes.</u>	1	No	1,950.00	1,950.00
	B1-0B-IDU-16, of air flow rate 228 L/s and total cooling coil load of 2.8 kW				
B	<u>HOOD MAKEUP AIR UNIT</u> <u>Hood Makeup Air unit, DX type, complete with, horse shoe heat pipe, cooling coil, condensate pan, fresh air fan, starters, filters, vibration isolator, necessary valves, Expansion kit, control panel, insulations and all necessary accessories and connections as per drawing, details and under Specification. Additionally, the VRV condensing unit shall comprises of multiple compressors, condenser coil, condenser fans, as per the specifications. The cost should include power connections between the isolator switch and the outdoor unit, spring isolator for the outdoor unit and concrete base with all the accessories, control and electric connections.</u>	1	No	97,883.00	97,883.00
	B1-0B-HMU-01 of air flow rate 730 L/s and total coil load of 14.8 kW				
C	B1-RF-CU-01 (VRV) total cooling coil load of 14 kW	1	No		
D	<u>REFRIGERANT PIPE INSTALLATION</u> <u>Supply, install and test refrigerant copper piping for interconnecting the VRF indoor and out door units as per the AC equipment capacity complete with pipe protection and insulation as per the supplier's recommendations and specification. The below pipe sizes are indicative, contractor shall verify them as per selected vendor.</u>	111	m	-	Included in Units
	Ø 6.40 mm				
	Ø 9.52 mm				
E		40	m	-	Included in Units
				QR.	99,833.00



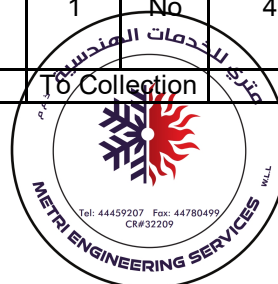
Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
	<u>REFRIGERANT PIPE INSTALLATION (Cont'd...)</u>				
A	Ø 12.7 mm	70	m	-	Included in Units
B	Ø 15.90 mm	45	m	-	Included in Units
C	Ø 19.05 mm	72	m	-	Included in Units
D	Ø 41.3 mm	35	m	-	Included in Units
	<u>DRAIN PIPING</u>				
	<u>Supply and installation of PVC-U drain pipe insulated with 19mm fibre glass pipe insulation.</u>				
E	Ø 20mm	138	m		
	<u>EXHAUST FANS</u>				
	<u>Supply Installation, testing & Commissioning of Low noise application exhaust fans, complete with motor, statically or dynamically balanced die-cast-in-aluminium impeller; rolled steel channel supporting frames, anti-vibration mountings, acoustic lining enclosure etc., complete as specified and indicated on the drawings, equipment schedule and technical specification of following capacities:</u>				
	<u>KITCHEN EXHAUST FAN</u>			GREENHECK INDIA	
	<u>Fan shall be listed in accordance with UL 762 or equivalent and suitable for commercial cooking</u>				
F	Exhaust Air Fan EF-B1-RF-02 of 807 L/s, Fire rated 400°C for 2 Hours	1	No	12,747.00	12,747.00
	<u>Supply Installation, testing & Commissioning of ELECTRO STATIC PRECIPITATOR as noted below</u>				
	<u>Constant flow ESP unit with pre-filter, electrostatic precipitator and carbon filter for kitchen hood exhaust, Floor mounted type unit, etc. complete as specified and indicated on the drawings, equipment schedule, and technical specification of following capacities, Airflow 807 L/s</u>			FREGO CHINA	
G		1	No	19,500.00	19,500.00
	<u>CAR PARK EXHAUST FAN</u>			GREENHECK INDIA	
H	Exhaust Air Fan EF-B1-0B of 1292 L/s, Fire rated 250°C for 2 Hours		No	6,080.00	6,080.00
	To Collection			QR.	38,327.00



Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
	AIR CONDITIONING DUCTWORK INSULATION <u>Supply, Fabrication and Installation of fiber glass board insulation material of 48 kg / m³ density, thermal conductivity limited to 0.037 W/m/°C and wrapped with reinforced aluminium foil and covered with high quality canvass cloth 200 gm/m³ soaked in weather proofing compound and sealed with vapor barrier seal (as applicable) as per technical specification and local standard.</u>				
A	For Concealed ductwork of 25mm thick insulation	1294	m2	60.00	77,640.00
B	For External ductwork of 50mm thick insulation	405	m2	90.00	36,450.00
	ACOUSTIC LINING <u>Acoustic Lining shall be of 15mm thick Mineral wool and faced to minimize fragmentation and fibre fly. The ductwork surfaces must be thoroughly cleaned and the lining fixed by an approved adhesive over the whole area to be lined.</u>				
C		150	m2	45.00	6,750.00
	KITCHEN EXHAUST DUCT <u>Kitchen exhaust ducts shall be constructed of and supported by carbon steel not less than 1.52 mm in thickness or stainless steel not less than 1.21 mm in thickness as per NFPA 96 caluse 7.5.1.1. Kitchen ventilation duct material shall be with fire rating equivalent to the enclosure wall being penetrated - 2 hour.</u>				
D		78	m2	250.00	19,500.00
	KITCHEN EXHAUST DUCT ACCESS PANEL <u>Supply and installation of Access panel for the Kitchen exhaust duct shall be constructed by stainless steel not less than 1.21 mm in thickness as per NFPA 96 caluse 7.5.1.1. Access panels shall have a gasket or sealant that is rated for 815.6°C (1500°F) and shall be greasetight as per NFPA 96 caluse 7.4.3.2. fixing position with duct work as per drawings, equipment schedule and technical specification of the following sizes.</u>				
E	600 x 300	4	No's	250.00	1,000.00
F	600 x 250		No's	250.00	750.00
				QR.	142,090.00

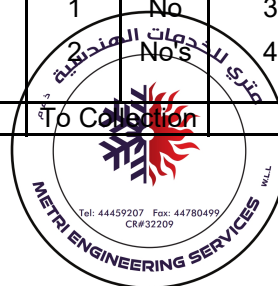


Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
	<u>FIRE DAMPER</u>				
	<u>Supply, installation, testing and commissioning of Fire damper as per technical specification and local standards and as shown in the drawings including all accessories; fix in position with ductwork of following sizes.</u>				
A	250x100	2	No's	347.00	694.00
B	200x100	2	No's	330.00	660.00
C	150x150	8	No's	277.00	2,216.00
D	150x100	1	No	277.00	277.00
E	200x150	2	No's	287.00	574.00
F	250x150	1	No	297.00	297.00
G	200x200	1	No	294.00	294.00
H	450x500	1	No	416.00	416.00
J	600x500	1	No	455.00	455.00
K	300x150	1	No	306.00	306.00
L	300x200	1	No	315.00	315.00
M	350x200	2	No's	325.00	650.00
N	450x350	1	No	376.00	376.00
P	400x300	1	No	356.00	356.00
Q	700x400	1	No	446.00	446.00
R	600x450	3	No's	444.00	1,332.00
S	550x400	2	No's	410.00	820.00
T	550x450	3	No's	432.00	1,296.00
U	900x700	2	No's	480.00	960.00
V	900x650	2	No's	571.00	1,142.00
W	600x300	1	No	400.00	400.00
			To Collection	QR.	14,282.00

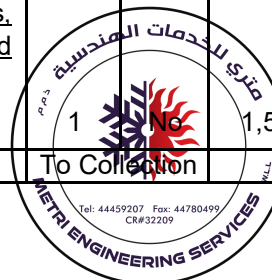


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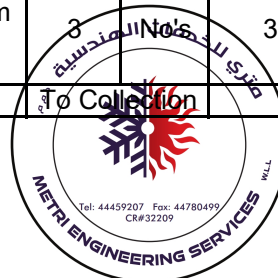
Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
	<u>VOLUME CONTROL DAMPER (CONT'D...)</u>				
A	450x350	2	No	235.00	470.00
B	500x200	2	No's	202.00	404.00
C	500x250	3	No's	214.00	642.00
D	500x300	1	No	219.00	219.00
E	500x350	3	No	243.00	729.00
F	500x400	2	No's	246.00	492.00
G	550x250	1	No	221.00	221.00
H	550x300	2	No	226.00	452.00
J	550x400	3	No	256.00	768.00
K	550x450	2	No	260.00	520.00
L	600x250	4	No	217.00	868.00
M	600x300	4	No's	222.00	888.00
N	600x450	2	No	253.00	506.00
P	650x400	1	No	288.00	288.00
Q	650x450	1	No	294.00	294.00
R	700x350	1	No's	294.00	294.00
S	750x350	3	No	305.00	915.00
T	800x300	2	No	276.00	552.00
U	800x500	2	No's	369.00	738.00
V	900x150	3	No	242.00	726.00
W	900x650	4	No's	453.00	1,812.00
X	900x700	2	No	456.00	912.00
Y	1000x300	1	No	308.00	308.00
Z	1000x650	2	No's	482.00	964.00
		To Collection		QR.	12,253.00



Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
	<u>EXTRACT AIR LOUVERS</u> <u>Supply, installation, testing and commissioning of Exhaust air louver with powder coated finish to approved color, including insect screen / bird guard, louver as per specification.</u>				
A	700x500	1	No's	397.00	397.00
B	1650x300	1	No	636.00	636.00
	<u>SOUND ATTENUATOR</u> <u>Supply, installation, testing and commissioning of Sound attenuators, including all accessories complete as per technical specification, local standards and as indicated on the drawings; jointing to ductwork (minimum length 1200mm) of following sizes in mm.</u>				
C	550 x 250	1	No	1,883.00	1,883.00
	<u>CROSS TALK ATTENUATOR</u> <u>Supply, installation, testing and commissioning of Cross talk attenuators, including all accessories complete as per technical specification and local standards. The size of the cross talk attenuator shall be followed as per the drawings.</u>				
D	800 x 300	1	No	900.00	900.00
	<u>MOTORIZED DAMPER</u> <u>Supply, installation, testing and commissioning of louver type motorised dampers confirm to SMACNA/NFPA in duct work with access door, actuator, matching flanges, gaskets & bolts, etc.. The item shall be complete including all the fittings and accessories as detailed in drawings and technical specifications and local standards. The actuator shall be ON/FF type. The actuator shall be capable of accepting 240 V AC supply, the controller shall be BMS ready and communicable to BACNET protocol with communication port. The item should be complete in all respect including all fittings, supports and accessories required for its complete installation and proper functioning as per standards, equipment schedule, technical specifications and details in drawings of following sizes.</u>				
E	500x400	1	No	1,500.00	1,500.00
		To Collection		QR.	5,316.00



Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
	<u>MOTORIZED DAMPER (CONT'D...)</u>				
A	600x300	1	No	1,500.00	1,500.00
B	600x450	2	No	2,500.00	5,000.00
C	600x500	2	No	2,640.00	5,280.00
	<u>FLEXIBLE DUCT</u>				-
	<u>Supply, installation and testing of Flexible duct not exceeding 1.5m, as per technical specification and local standard.</u>				-
D	Ø 200mm	48	m	50.00	2,395.00
E	Ø 150mm	78	m	40.00	3,120.00
	<u>BELL MOUTH WITH WIRE MESH</u>				-
	<u>Supply, fabrication, and installation of bell mouth with wire mesh, complete with necessary flanges, supports, and fixing accessories. The wire mesh shall be of aluminium with a minimum mesh size of [8-13] mm. All components to be as per project specifications and Tender drawings. The size of the bell mouth with wire mesh shall be followed as per the drawings.</u>				-
F	<u>AIR INLETS AND OUTLETS</u>				
	<u>Supply, installation, balancing/testing of Supply, return and exhaust air grilles and diffusers along with insulated plenum boxes (as applicable) as specified in the technical specification, local codes and standards and drawings, extruded aluminium construction with powder coated finish to approved colour; complete with supports and flexible connections; fixing in position including connections to ductwork as per standards, equipment schedule, technical specifications and details in drawings of following sizes.</u>				
	<u>Supply air linear diffuser with dampers, sizes as follows</u>				
G	2Slot Linear Diffuser, 1000mm length and 20 mm wide	16	No's	300.00	4,800.00
H	3Slot Linear Diffuser, 1000mm length and 25 mm wide	3	No's	300.00	900.00
				QR.	18,065.00



Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
	<u>Collection:</u>				
	Page 2/8/1				265,000.00
	Page 2/8/2				-
	Page 2/8/3				163,000.00
	Page 2/8/4				99,833.00
	Page 2/8/5				38,327.00
	Page 2/8/6				180,000.00
	Page 2/8/7				142,090.00
	Page 2/8/8				14,282.00
	Page 2/8/9				41,000.00
	Page 2/8/10				12,253.00
	Page 2/8/11				7,684.00
	Page 2/8/12				5,316.00
	Page 2/8/13				18,065.00
	Page 2/8/14				46,472.00
	Page 2/8/15				57,500.00
	Page 2/8/16				-
	Page 2/8/17				-
	<u>TOTAL FOR HVAC INSTALLATIONS</u>			QR.	1,090,822.00

